

A NOT_MEASURED REGISTER FOR A WITHDRAWN KV-SELECTIVE SERVING STUDY: SCOPE, FROZEN VALUES, AND A GUARDED RERUN

Anonymous authors

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ABSTRACT

We document the withdrawal of a frozen repeated-query serving study whose evidence did not support its original cost-quality interpretation. The note contributes three reporting instruments: scoped statements for each surviving claim, a **NOT_MEASURED register** that is complete relative to an author-enumerated claim ledger (not an independent ground truth; Section 4), and frozen values with per-row scope labels. Pattern-enumerated checks make violations within their declared phrasing universe re-runnable, while leaving coverage outside that universe explicitly open.

Static audit identified two application-level defects that had previously been read as hardware boundaries: a thread-local grad-mode leak in the full-context worker (≈ 2 lines, fixed and re-measured) and an argument mismatch in a vendored KV-selector (≈ 6 – 10 lines, statically diagnosed only). The guarded concurrent rerun (D1) establishes feasibility at $L=32k$: at concurrency $G=1$ the peak is 29.31 GB, decode is finite, and no out-of-memory failure occurs. At 16k, the measured pre-guard peak is 139.2 GB, but its attribution to the missing guard remains inspection-grade because no guarded concurrent rerun was produced. The 16k rerun, KV-selector arm, and adapter-matched arm therefore remain NOT_MEASURED. The earlier “infeasible environment” interpretation is withdrawn, not replaced by a positive serving claim.

1 INTRODUCTION AND SCOPE

An archived study of multi-query KV-selective serving was withdrawn after its load-bearing claims could not be supported: the worker’s GPU allocation was exhausted, the declared adapter was unrecoverable, and concurrent serving aborted. We report both the withdrawal and the *reporting apparatus* developed to delimit what remains measurable and citable.

Contributions. Figure 1 summarizes four outputs: (a) scope statements for each surviving claim with re-runnable, pattern-scoped checks (Section 3); (b) a NOT_MEASURED register complete only relative to the author-enumerated ledger (Section 4); (c) frozen values labelled by concurrency, estimand class, and status (Section 5); and (d) a documented withdrawal in which a guarded rerun resolves the apparent memory ceiling at one length but not the other (Section 6). All measurements are scoped to one H20-3e GPU (141 GB (Yang et al., 2025)), Qwen3-8B in `bfloat16`, SDPA, and no adapter unless stated. Multi-GPU, multi-day, and 1M-token concurrent serving are outside scope.

What this note does not establish. It does not reinstate the archived cost-quality conclusion, provide a concurrent throughput or latency result, or verify adapter-arm identity; these omissions are recorded as register rows and a binding consumption discipline (Section 7). The archived study used short internal identifiers; Table 1 defines the ones the argument needs and Table 8 (Appendix F) the full ledger, so the note reads from the PDF alone.

2 RELATED WORK AND POSITIONING

Measurement and reporting methodology. The closest prior work is not a serving system but the literature on how empirical claims should be measured and reported. Systems benchmarking

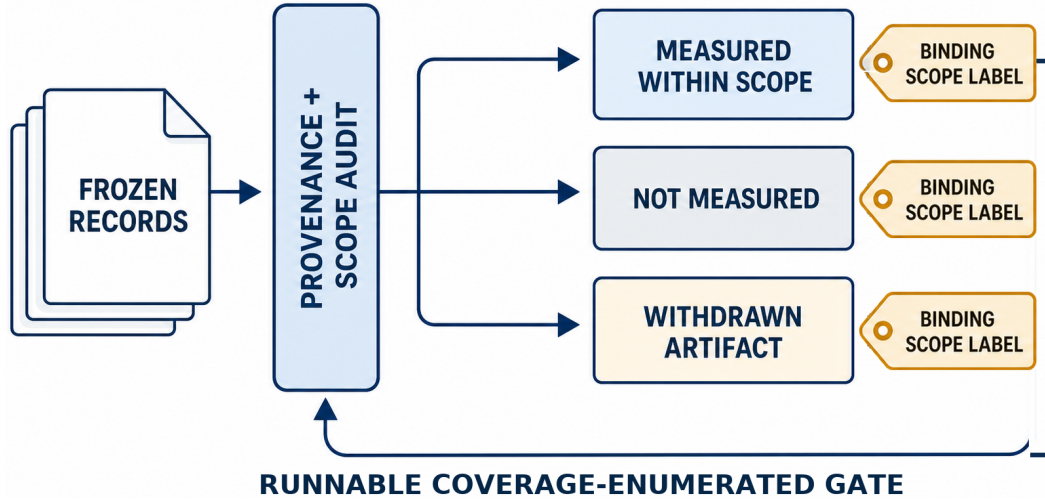


Figure 1: The three instruments this note contributes, and how they compose. Frozen records pass one provenance-and-scope audit that assigns exactly one of three dispositions; each disposition travels with a scope label documenting how the value may be cited; and a runnable, coverage-enumerated (pattern-scoped) gate re-checks the manuscript itself against that discipline (Section 3.1). The diagram is methodological and contains no quantitative evidence.

Table 1: Core notation. Arm names describe *where* a query’s context is reconstructed from; the full ledger-ID glossary is Table 8 in Appendix F.

Term	Definition
CoMem	The archived cached-memory serving system; cached arm.
$j0$ ($j=0$)	The <i>full-depth</i> arm: nothing cached; selected chunks replay all layers.
$j12$	The flagship <i>cached</i> arm ($j=12$ states + LoRA (Hu et al., 2022)).
B-only	The full-context dense baseline <i>without</i> the (unrecoverable) LoRA arm; every staircase cell is B-only.
S	Per-query CPU cost of chunk- <i>selection</i> ; algorithmic estimate, not a frozen-corpus measurement.
G	Concurrency; or the generated length in the archived decode model (as labelled per row).
Q^*	Break-even query count of the cached arm vs the full-depth arm (Eq. 1).
C/D/S/R-family IDs	Stable claim IDs of the archived ledger (Table 2 and Appendix F).

names how performance results mislead (Hoefer & Belli, 2015) and how bias can invert a conclusion drawn from careful experiments (Mytkowicz et al., 2009) — our failure mode, since the “capacity boundary” was a property of the measurement surface. Machine learning has adopted budget-aware reporting (Dodge et al., 2019), reproducibility checklists (Pineau et al., 2021), artifact evaluation (Krishnamurthi & Vitek, 2015), and structured documentation templates (Gebu et al., 2021; Mitchell et al., 2019). Two strands bear on a withdrawal note: pre-registration, which fixes analysis scope before the result is known (Nosek et al., 2018), and non-replication reporting (Ferrari Dacrema et al., 2019; Kapoor & Narayanan, 2023). Closest in spirit, assurance-case notation (the Goal Structuring Notation; Kelly & Weaver, 2004) already makes *undeveloped goals* — claims an argument states but does not yet establish — explicit in a structured tree. That idea is not new.

What a NOT_MEASURED register adds over a Limitations section. Given that making unestablished claims explicit predates this work (GSN (Kelly & Weaver, 2004)), the register is a *narrow, machine-checked instance* of that idea, not a new argument formalism. A Limitations section is free-form prose, and dropping an item need not trigger any failure; datasheets and model cards fix the *template*, checklists fix the *questions*. The register adds three machine-checkable properties (Section 4, Table 3, Section 3.1): a *complete inventory relative to a declared universe*, *per-row binding labels* (concurrency, estimand class, status) that forbid re-citing a value outside the scope that produced it, and a *machine-checked* scope predicate a reviewer runs, exiting non-zero on violation.

Technical background for the withdrawn study. For context only, and contributing nothing here: the archived study sat among IO-aware exact attention (Dao et al., 2022), token- and page-level

KV-cache management (Zhang et al., 2023; Xiao et al., 2024; Adnan et al., 2024; Kwon et al., 2023; Ge et al., 2024), and length-scaled long-context quality probes (Hsieh et al., 2024; Li et al., 2023); its statistics use the exact conditional paired test (McNemar, 1947), percentile bootstrap (Efron & Tibshirani, 1993) and Wilson (Wilson, 1927) intervals. The memory-inflation behind the withdrawal is autograd bookkeeping: outside an inference-mode guard the framework retains per-layer activations for a backward pass that never happens (Paszke et al., 2019).

3 ADJUDICATED BOUNDARIES AND THEIR ENUMERATED (PATTERN-SCOPED) CHECKS

3.1 SERIAL-LEDGER DECOUPLING IS SCOPED, AND THE SCOPE IS MACHINE-CHECKED

The same-term cancellation in the archived cost accounting applies to the **serial ledger** only, and we make **zero** wall-clock or concurrency-cancellation claim from that scoping. The only concurrent wall-clock ever measured on this stack (full-context, no adapter and no selector; Section 6) carries no selection-cost S term, so there is no concurrent critical path on which an S -cancellation could be asserted; the S -removal is licensed **only** as a serial-ledger same-term cancellation ($S_0=S_c$ on both serial-ledger sides). A reviewer re-runs the enumerated scope audit `python3 bundle/scanners/audit_p0_l_critical_path_scope.py --note bundle/note/main.tex` (0 GPU, standard library only); it exits 0 only when all four gates hold. The predicates, as implemented post-D1, with their arm restrictions:

- **G1 (required).** An explicit serial-ledger-only scoping statement is present, matched on the literal tokens `serial-ledger`, `serial ledger`, or `same-term cancellation`.
- **G2 (required, arm-restricted).** A concurrent-serving NOT_MEASURED declaration is present, matched on `not_measured`, `not measured`, or `unmeasured`. Post-D1 that declaration is *arm-restricted*: it governs the KV-selector ($j0$) arm, the adapter-matched arm and the 16k full-context arm, and **not** the full-context arm at $L=32k$, which Table 2 declares MEASURED. The gate checks such a declaration exists, not that every arm is unmeasured, so the two statements are consistent.
- **G3 (required).** The withdrawn reading is not re-asserted: four phrase patterns are banned in non-negated sentences, including `permanent (memory|capacity|hardware) bound and verified in code`. The script’s original docstring also required the re-attribution to be called *suspected*; that gloss is superseded because D1 measured it at 32k. The implemented predicate is unchanged.
- **G4 (banned).** No concurrent-critical-path S -cancellation claim and no concurrency-throughput reading appears, matched against a published enumeration of eleven phrase patterns in non-negated sentences.

The gate has two published limits. First, it enumerates phrasings: an earlier revision passed G4 while a throughput-style reading sat in prose outside the enumeration; that reading was removed, the enumeration extended, and the failure kept as a runnable red-team fixture (`bundle/scanners/redteam/`) so a reviewer confirms the gate now rejects it. Second, *verbatim code spans are exempt* from matching — which is what lets this section print the banned literals as specification, not assertion; prose is never exempt, so no enumerated place within the declared phrasing universe remains to hide a claim. A reviewer who finds another uncovered phrasing has falsified the gate’s coverage, not merely disagreed. Bounded by its enumeration, the gate is falsifiable exactly because it is.

3.2 THE MACHINE-CHECK IS A TRANSCRIPT, NOT A SLOGAN

“Machine-checked” is a concrete run, not a prose promise: Table 5 in Appendix B lists every scanner that shipped with this note, what it targets, the exit code on the frozen text at compile time, the counts it emits, and — for the scope and register gates — a red-team fixture that must *fail*, proving the check has discriminative power rather than always passing. A reviewer re-runs any row against `bundle/` (0 GPU, stdlib only).

3.3 ADAPTER IDENTITY IS PARTIAL AND PROVISIONAL

The shared-flagship-LoRA arm identity was **provisional**: the declared matched adapter (dd09cd17...0536f) could not be retrieved or reproduced, so any dependent specialization claim is provisional. Two facts are separated so each is independently verifiable. **FACT-A**, the measured serving record’s own manifest is authentic: `sha256sum -c SHA256SUMS.txt` run *inside* `bundle/p1_8_serving/` reproduces **24/24 files OK** in seconds — the record’s original manifest, a different file from the bundle-wide `bundle/SHA256SUMS.txt` that covers every other shipped artifact. **FACT-B**, the adapter weights are not recoverable: zero LoRA weight files (`.safetensors`, `.bin`, `.pt`) exist anywhere in the bounded release tree, and dd09cd17 is governed by a withdrawn disposition. FACT-A does not imply FACT-B’s negation: an authentic record of a run is not a recoverable copy of the weights that ran. This note therefore claims **no** verified adapter-dependent specialization; re-run `python3 bundle/scanners/verify_p0_2_adapter_provenance.py --bundle bundle/` to confirm both.

3.4 LONGEVAL IS DIRECTION-ONLY

On the LongEval line-retrieval suite (Li et al., 2023), per-cell Wilson 95% intervals (Wilson, 1927) separate in the same direction at every tested length. But **no paired test is claimed and no LongEval value is citable from this note**: per-item raw outputs were never retained, so the archived pooled statistic is a descriptive comparison between independent aggregates, not a paired significance result on matched items. The paired re-run is register row D6 of Table 2; until it exists, the LongEval evidence is directional only.

4 THE NOT_MEASURED REGISTER

Table 2 is the register, and **its universe is declared, not rhetorical**: the rows enumerate the load-bearing claims of the archived claim ledger at closure — the C-numbered serving/quality claims, D-numbered decisive-evidence debts, and S-numbered soundness-deficit classes. Completeness is asserted *relative to that enumerable, shipped ledger*, not to every statement a reader might form, and is falsified by exhibiting a ledger item with no row. The limit is that the ledger is *author-constructed*: the register certifies no *ledgered* claim lacks a row, but cannot prove none was ever omitted (third-party recovery is the narrowing check we have not run); and the exact-set *semantic* check is clue-scoped, not a general guarantee — each registered ID appears and every canonical semantic clue in the text after it belongs to that same ID (a swap that moves one ID’s meaning onto another’s row is caught by the shipped mutation fixture), but a substitution borrowing no canonical clue, or one outside the per-ID text window, is unseen. A new failing mutation falsifies one of these bounds, as the red-team fixtures do.

Exact-set completeness is machine-checked. Table 2 is *generated* from the checked-in single source `tools/register_ledger_check.py` (LEDGER + REGISTER + DECLARED_MERGES), and every row carries an explicit **Ledger-ID** column that spans the full C/D/S closure universe: C-numbered claims retained, D-numbered decisive-evidence debts, S-numbered soundness-deficit classes. The generator and the checker enforce the same invariants — every ledger ID has a row, no row adds an unknown ID, and any many-to-one merge (e.g. $S1/S3/S6 \rightarrow$ one row) is declared and counted — so deleting a row (e.g. omitting D4) or diverging a status canonical token makes the build fail non-zero, rather than silently passing (Section 3.1).

One cross-scope reference, labelled as such. The guarded *serial* staircase records 22.30 GB at 16k (C7, serial ledger, $G=1$; Table 3). For the D1-16k row of Table 2 that value is a **serial cross-scope reference**, not a concurrent measurement of it and not an independent confirmation of it. Its role is bounded and stated in Section 6: same model, dtype, attention backend and GPU as the pre-guard concurrent 16k observation, hence a same-hardware comparison point for the guard mechanism.

5 FROZEN MEASURABLE VALUES, WITH PER-ROW BINDING SCOPE

The values in Table 3 are citable **only** with the scope printed on their own row. There is no table-wide scope, because the rows are heterogeneous by design: C13’s replicate row is a $G=512$ figure, not a

Table 2: The NOT_MEASURED register, machine-generated from the checked-in single source (tools/register_ledger_check.py). Each row binds Ledger-ID entries — spanning the C (claim retained), D (decisive-evidence debt) and S (soundness-deficit) families — to their arm and scope, canonical status, and closing measurement. Statuses come from one vocabulary shared with Table 3: MEASURED, DERIVED, ESTIMATE (algorithmic, not measured), EXTRAPOLATED (outside the measured range), NOT_MEASURED, CLOSED_WRITING (closed by a writing-only declaration, not empirically), NOT_ADJUDICATED (reported for completeness, not citable as a prediction). The last two are printed because the checker fixes the vocabulary, not this table; no row here carries them. The D4 row and the S1/S3/S6 merged row instantiate the many-to-one semantic formalization.

Ledger ID	Ledger item	Arm / scope	Status	What would close it
C4	serial-cost ledger: CoMem/j0 serving cost terms	family C; serial cost	MEASURED	MEASURED from frozen 24-file manifest (p1_8_serving); citable within serial-cost scope
C7	memory slope vs time-power-law exponent (B-only capacity staircase)	family C; serial capacity; $G=1$	DERIVED	co-measured memory and prefill-time record at the same operating point
C9	RULER macro accuracy gap and bootstrap	family C; quality; macro over 15 cells	DERIVED	item-level paired McNemar on the retained raw outputs
C13	Q^* crossover estimate class	family C; serial cost; crossover	ESTIMATE	measured per-query margin at $G=1$ (not an estimate)
D1	concurrent full-context serving @ $L=32k$ (grad-mode fix-and-rerun)	family D; full-context; concurrent $G \in \{1, 4\}$; $L=32k$	MEASURED	MEASURED (D1 fix-and-rerun at $L=32k$, no OOM; Table 4; throttle curve remains open)
D1-16k	concurrent full-context serving @ $L=16k$ re-run	family D; full-context; $L=16k$	NOT_MEASURED	NOT_MEASURED: re-run the guarded full-context arm at $L=16k$; the current 16k peak is inspection-grade cross-scope reference only (Section 6)
D2	adapter-source recovery / rebuild	family D; adapter; serial and concurrent	NOT_MEASURED	recover or rebuild the declared adapter, then re-run
D3	KV-selector ($j0$) finite fetch	family D; selector; concurrent	NOT_MEASURED	repair <code>create_causal_mask</code> argument mismatch (three call sites), then re-run
D4	adapter-identity table completeness	family D; adapter; writing declaration	CLOSED_WRITING	CLOSED_WRITING: closed by a writing-only declaration, not empirically (this table) — completeness is relative to the enumerated ledger
D5	adapter-identity per-row binding caveat	family D; adapter; per-row scope	CLOSED_WRITING	CLOSED_WRITING: every identity value carries a per-row binding caveat (this table), not a verified equality
D6	LongEval per-item (paired) re-run	family D; quality; LongEval; direction-only currently	NOT_MEASURED	retain per-item outputs and run paired McNemar; needs GPU and a scorer environment
S1/S3/S6	single-arm isolation of the scoped mechanism	family S; any arm; serial	NOT_MEASURED	single-arm run with co-measured cost and quality

Table 3: Frozen measurable values with per-row binding scope. “ G ” is the row’s concurrency or generated-length setting (— where concurrency does not apply, as for accuracy estimands). Status tokens are the canonical vocabulary of Table 2: MEASURED (read from a frozen record), DERIVED (computed from measured values by the stated estimator), ESTIMATE (algorithmic only), NOT_ADJUDICATED (reported for completeness, not citable as a prediction). Rows are grouped by claim family. No cross-row aggregation and no concurrent-serving inference is licensed by any row.

ID	Quantity	Value (units)	G	Estimand class	Status
C4	CoMem write @128k, one-time	6.65 s	1	serial cost	MEASURED
C4	$j0$ index build @128k, one-time	0.022 s	1	serial cost	MEASURED
C4	CoMem cache read, per query	0.68 s/q	1	serial cost	MEASURED
C4	$j0$ read, per query	0.94 s/q	1	serial cost	MEASURED
C4	fetch, per query (CoMem / $j0$)	0.0014 / 0.00002 s/q	1	serial cost	MEASURED
C7	B-only full-context peak at 8/16/32/64/128k	18.80/22.30/29.31/43.34/71.38 GB	1	serial capacity	MEASURED
C7	B-only cache peak @128k	18.29 GB	1	serial capacity	MEASURED
C7	full-context prefill time at 8/16/32/64/128k	1.28/2.98/7.75/22.26/72.74 s	1	serial cost	MEASURED
C7	growth exponent of prefill time in L	1.46 (dimensionless)	1	serial fit, 5 points	DERIVED
C9	RULER macro accuracy gap	3.12 pp	—	quality, macro over 15 cells	MEASURED
C9	cell-level 95% percentile bootstrap CI	[1.69, 4.68] pp	—	quality, cell level	DERIVED
C9	item-level discordant counts $b/c/n$	83 / 1 / 84	—	quality, pooled item level	MEASURED
C9	exact two-sided McNemar p	8.79×10^{-24}	—	quality, pooled item level	DERIVED
C9	pooled accuracy gap	5.47 pp	—	quality, pooled	MEASURED
C13	paired per-query margin @128k	0.0404 s/q	512	crossover input	MEASURED
C13	Q^* @128k (aggregate grid)	25.8 queries	1	crossover, model-implied: Eq. 1 with inputs in Table 6	DERIVED
C13	Q^* @32k	8.4 queries	1	crossover, model-implied: Eq. 1; inputs not separately printed	DERIVED
C13	Q^* @1M	180.2 queries	1	crossover beyond the measured staircase ($\leq 128k$); model-implied	EXTRAPOLATED
C13	Q^* @128k, replicate	164.2 queries	512	crossover, replicate	DERIVED
R142	full-context write model $W(L)$	$4.432 \times 10^{-5} L + 0.775$ s	1	in-sample fit, 3 points	NOT_ADJUDICATED

$G=1$ serial one, and C9’s rows are accuracy estimands, not cost figures. An earlier version carried a universal “every value is a serial $G=1$ figure” preamble that its own C13 row falsified. Rows must not be combined into a serving $\times$ quality metric.

Estimators, so every derived row is reproducible from the printed inputs.

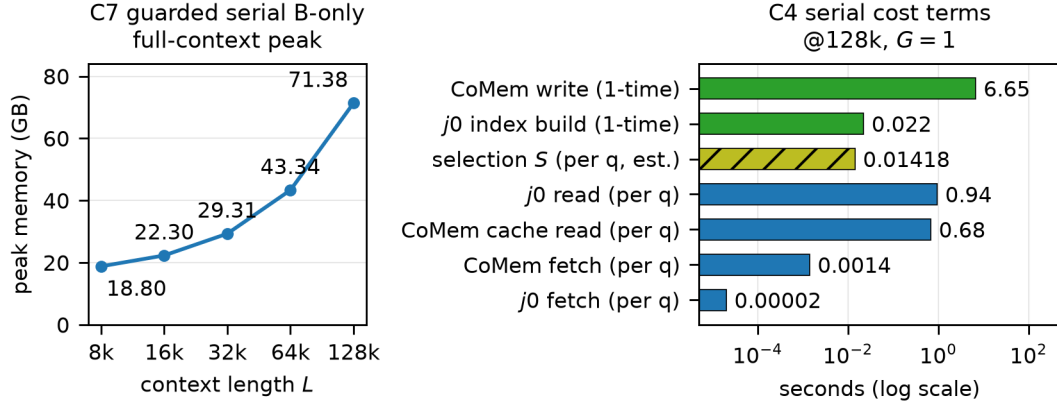


Figure 2: Frozen serial-ledger profiles, plotted from the bundled datacards behind Table 3. **Left:** guarded B-only full-context peak memory rises monotonically with context length (bundle/staircase/fullctx_serving.json); this is the guarded serial surface, and the pre-guard concurrent observations of Section 6 are not on it. **Right:** the @128k, $G=1$ cost terms on a log axis, split into one-time and per-query classes, the hatched bar being the unmeasured algorithmic estimate S ; each tick label carries its class, (1-time) for a cost paid once per store and (per q) for one paid per query. The log axis is needed because the terms span five orders of magnitude, and the one-time/per-query split is what keeps the 0.022 s index build from being read as a per-query selection cost. Both panels are scoped serial references, never concurrent-serving or capacity claims.

- **C9.** On the discordant counts $b/c/n$ printed in Table 3, the two-sided exact conditional McNemar test gives $p = 2 \sum_{k=0}^{\min(b,c)} \binom{n}{k} 2^{-n}$, so those counts alone reproduce the printed p . That test is at the *pooled item level*, whereas the interval is a percentile bootstrap at the *cell level* (15 RULER cells, seed 123, $n_{\text{boot}}=20000$) around the macro gap. **No test is claimed for the macro gap**, and the pooled gap is a same-direction but distinct estimand rather than a second estimate of it. Figure 3 (Appendix E) plots the two intervals this note cites; neither includes zero.
- **C13.** With per-query cost $c_{\text{arm}}(G) = \text{fetch} + \text{read} + \text{decode}(G)$,

$$Q^*(L) = \frac{W_{\text{CoMem}}(L) - I_{j0}(L)}{c_{j0}(G) - c_{\text{CoMem}}(G)}. \quad (1)$$

The numerator differences the two *one-time* costs; the denominator is the mean paired per-query margin, defined only where positive. Q^* grows with G because decode does *not* cancel between the arms, so its values at two concurrency settings are *different quantities* rather than a contradiction; and S does *not* enter Eq. 1, so every input is measured. Table 6 (Appendix D) carries the inputs, the quotients and the per-row provenance flag, including the superseded face whose denominator is not reproducible from the printed components.

- **C7.** The three slopes of Table 7 (Appendix D) are *different quantities* and must not be interchanged: 1.46 is the least-squares slope of $\log t$ on $\log L$ over the five *prefill times*; the same estimator on the *peak-memory* series gives 0.48, and peak memory is near-affine in L at 0.438 GB per 1k tokens.
- **S versus C4’s selection-side terms: different quantities, not a discrepancy.** S is the *per-query* CPU cost of scoring chunks (Table 7), an algorithmic estimate on representative chunks rather than the frozen corpus. C4’s “j0 index build” (0.022 s @128k) is the *one-time* $O(L)$ index construction over the whole store, entering Eq. 1’s numerator beside the one-time write rather than any per-query term; C4’s “j0 read” (0.94 s/q) is the per-query 36-layer replay. The three sit at different decomposition granularity and are not expected to reconcile numerically; Figure 2 (right) shows the split.

6 THE D1 CONCURRENT FIX-AND-RERUN, AND WHAT IT DOES NOT COVER

Two application-level defects, one fixed and measured, one only diagnosed. The concurrent experiment was previously reported as blocked by the environment; static audit of our own measurement code found two defects with different evidential status instead. (i) A *thread-local grad-mode leak*:

Table 4: The D1 concurrent full-context measurement at $L=32k$, after the two-line inference-mode guard. Single H20-3e (141GB), Qwen3-8B bfloat16, SDPA attention, store $L=32k$, read sample 8k, task niah_multikey_1, 8 queries, $R=3$ rounds, seed 42, no adapter, no selector. Wall-clock is *per request* and includes queue wait under contention. Dispersion is the coefficient of variation, which is what the driver retained. Source bundle/d1/e2e_concurrent_kvsel.json; applied diff bundle/d1/D1_FIX.diff.

G	Peak (GB)	Median (s)	CV	n ($R \times G$)	Role of this row
1	29.31	7.71	0.031	$3 = 3 \times 1$	consistency check (see below)
4	70.73	30.63	0.001	$12 = 3 \times 4$	the concurrency datum
pre-guard 32k	none (abort)	—	—	1	no peak recorded; the 137.4 GB figure is analytical only, never measured
pre-guard 16k	139.2	—	—	1	cross-scope reference, inspection-grade vs serial 22.30 GB (see text)

the concurrent worker called the model outside any `torch.inference_mode()/no_grad()` scope that the serial baseline wrapped the same forward in. Because grad mode is thread-local, worker threads inherited autograd *on* and retained 36 layers of activations for a backward pass that never occurs (Paszke et al., 2019). The repair is ≈ 2 lines and **has been applied and re-measured**. (ii) A *framework-version argument mismatch*: the vendored QCMem selector calls `create_causal_mask` with keyword `inputs_embeds`, whereas installed `transformers` 4.54.1 names that parameter `input_embeds`; the ≈ 6 –10-line repair across three call sites needs no `transformers` ≥ 5.5 upgrade but is **statically diagnosed only, not re-measured**. The two stay separate rather than being summed into one “ ≈ 10 -line” blocker, because only the first is backed by a measurement.

What D1 measured. After the guard, the full-context arm at $L=32k$ runs end to end at both tested concurrency levels, with finite decode and no out-of-memory failure; every value is on its own row of Table 4, and the matched pair is *pre-guard abort* versus a *post-guard finite peak*. Two scope labels attach to the two pre-guard rows of that table. First, the pre-guard 32k job recorded **no peak** before aborting, so the tensor-size figure sometimes attached to 32k is *analytical*, from a superseded root-cause account, and never a measured peak. Second, the pre-guard 16k job *did* record a peak, on the same model, dtype, attention backend and GPU as the guarded serial surface — consistent with activations retained across all 36 layers. That 16k comparison is *inspection-grade*: no 16k concurrent re-run was produced, so the 16k row stays NOT_MEASURED (Table 2), even though its peak is a measured record. The earlier reading of D1 as a memory or infeasibility *boundary* is therefore withdrawn — at 32k the full-context concurrent state is feasible and the inflation was a property of our measurement code — while the KV-selector arm, the adapter-matched arm and the 16k re-run remain unmeasured. D1 closes exactly one register row.

Replication structure, stated rather than implied. The n column counts *per-request* samples: $n=12$ at $G=4$ is 3 rounds of 4 simultaneous requests sharing a queue and a device, so the effective independent replication is **3 rounds per level**. The driver retained only median, CV and n ; per-request wall-clocks were not persisted, so an IQR or range is not recoverable from the frozen record, and we do not manufacture one.

The $G=1$ row is a consistency check, not a second concurrency datum. At $G=1$ the driver submits one in-flight request through the same full-context forward path as the serial staircase: its peak equals the serial C7 32k entry of Table 3 exactly, and its median wall-clock agrees with the serial read time to within 0.6%. We read this as cross-surface agreement that the guarded surface is the same surface — **not** an independent second concurrency level; the withdrawal’s concurrency content rests on the $G=4$ row alone.

No throughput reading is offered. The two rows are peak-feasibility and wall-clock records. The $G=4$ to $G=1$ median ratio is ≈ 4 , but with one reference implementation, one operating point per level, 3 effective replicates and no controlled contention arm, the ratio identifies no mechanism; we make **no** concurrency-throughput, latency or scalability claim (a prior version printed the ratio at three significant figures and glossed it as a throughput result; that reading is removed, not re-disclaimed).

Quality debt carried forward (D2–D6, S1). D2, adapter source recovery, is an external artifact-access problem rather than a code fix. D6, the LongEval per-item re-run, needs GPU and a scorer

environment. D4, adapter identity in the archived tables, is closed here as a writing-only declaration (Section 3.3). S1, single-arm isolation, needs GPU; D3, the KV-selector ($j0$) finite fetch, is statically diagnosed but not re-measured.

Environment-boundary honesty. No failure was retried to burn: every failing job was stopped and reported, and all GPU consumption is in the shipped burn ledger (cumulative 1.82 GPU·h, “0 data” across five fires). The driving code logs `torch.cuda.max_memory_allocated()` before failing, so a pre-guard peak exists at 16k; the earlier 32k job predates that logging, which is why none exists there.

7 CONSUMPTION DISCIPLINE (BINDING ON RE-USE)

The following are required interpretations for any later text that re-uses values from this note; they are prohibitions on inference, not on discussion.

- **Do not** re-assert “selection is cheap or S -symmetric, therefore concurrent wall-clock cancels”: no concurrent selector critical path was ever measured.
- **Do not** read the LongEval gap as a paired significance result, and do not claim adapter-arm identity is verified.
- **Do not** read Table 4 as throughput, scalability or latency.
- Cite frozen values only with the scope printed on their own row (Table 3): no derived extrapolation, no combined serving $\times$ quality metric.
- Report failures loudly; do not re-fire a failed job to consume budget.

8 METHOD VALIDITY: CROSS-CASE TRANSFER

The instrument is not a description of this one withdrawal case: the read-only six-family scanner (`bundle/method/independent_claim_audit.py`, 0 GPU) also ran on an **independent project** — the S1 SparseForge manuscript (934 lines, a different group’s paper, different area) — striking **91 signal hits** across all six families (`audit_sparseforge.json`) under a different, *provenance-crisis-led* honesty structure. The same tool yields 68 hits on this note (withdrawal-dense, 28 WITHDRAWN), so the scan distinguishes failure-mode *structures*, not a “more hedging = more honest” counter; the per-family split of both runs is the last row of Table 5 (Appendix B).

The transfer shows the *scan* generalizes, not that any downstream evaluation score is comparable across projects. Two validity checks are reported in Appendix C: an *independent-annotator* recovery study (a reader blind to our ledger recovered all 14 register IDs at the paper’s semantics) and a *coverage-self-critique* of the six-family taxonomy, which is **not exhaustive** — the scanner is a *necessary* hygiene filter, not a complete-honesty certificate.

Layout step-width is NOT_MEASURED on held-out partitions. The mechanical formatting/layout revisions in this note’s history (float placement, collapse to the 9-page main-text limit, caption and font scaling) are NOT_MEASURED improvements: none was re-verified on an un-measured/held-out visual partition by an independent layout audit at every step. We claim only that the artifact’s content and scope gates hold; the layout step-width is not a measured gain and must not be cited as one until an independent visual audit passes on a frozen face.

9 CONCLUSION

The guarded rerun shows that the full-context arm at $L=32k$ is feasible at both tested concurrency levels, with finite decode and no out-of-memory failure (Table 4). This establishes that the earlier apparent environment or hardware boundary was, at that length, a measurement-code defect. The frozen serial values remain citable only within their row-specific scopes. The KV-selector arm, adapter-matched arm, and guarded 16k concurrent rerun remain unmeasured; the 16k attribution is inspection-grade, and the $G=4$ row is the sole concurrency datum, with 3 effective replicates. The contribution is therefore a reporting instrument rather than a serving result: a declared-ledger register, per-row scope labels, and an enumerated pattern-scoped gate whose uncovered phrasing remains criticizable. The instrument applies only where an enumerable claim ledger exists and cannot replace the missing measurements; it makes their status auditable within this record.

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A REPRODUCIBILITY AND ETHICS STATEMENTS

Every value citable in this note is verifiable against a shipped frozen record (checkable, in `bundle/` with `SHA256SUMS.txt` and per-item READMEs), and most derived quantities re-compute from their printed inputs with 0 GPU and only the standard library. Not every recovered artifact is end-to-end re-runnable — the declared adapter and the LongEval per-item outputs are unrecoverable — so “runtime” is not claimed where it does not hold. The note source is bundled at `bundle/note/main.tex` so the scope gate can run against the frozen text itself.

Artifact verification. An anonymized archive of `bundle/` (with its `SHA256SUMS.txt` and per-item READMEs) ships at the link on this manuscript’s blind submission; no artifact path here is specific to any author identity. A reviewer verifies the retrieved bytes by digest: the bundle manifest `SHA256SUMS.txt` has `dfa477175acb79497e59a0de7c3aad9de72bd72d2c30ba99f050d196d9567322` and `sha256sum -c SHA256SUMS.txt` inside the retrieved `bundle/` reproduces all shipped artifacts — including the manuscript source `bundle/note/main.tex` that the scanner registers (Section 3.1, Section 4) run against. The compiled PDF is produced deterministically from that source, so its digest is recorded in this revision’s build receipt and re-derived by recompiling; a reviewer needs only the standard library and 0 GPU.

- **Scanners** (G1–G4 scope, adapter provenance, grad-mode, register): see the transcript in Table 5 (Appendix B) and Section 3.1.
- **D1 measurement** (Table 4): `bundle/d1/e2e_concurrent_kvsel.json`, applied diff `bundle/d1/D1_FIX.diff`, gate verdict `bundle/d1/feas_verdict.json`; pre-guard evidence `bundle/oom/fullctx_16k_oom.log` (139.2 GB recorded) and `bundle/oom/fullctx_32k_oom.log` (abort) — records of the pre-fix surface, never capacity bounds.
- **Datacards** behind Table 3: `bundle/staircase/fullctx_serving.json` (C7), `bundle/ruler_ci_bootstrap.json` (C9), `bundle/qstar_dispersion_reanalysis.json` (C13; superseded by the aggregate grid), `bundle/s_estimate_iterbm25.json` (S), and `bundle/pl_8_serving/` (C4/C13, 24-file manifest).
- **Method-transfer audit**: `bundle/method/independent_claim_audit.py`, `audit_sparseforge.json`, `audit_ownnote.json`.
- **GPU accounting**: `bundle/burn/HONEST_BURN_*.md`.

Ethics. This work involves no human subjects, no personal or sensitive data and no model release. Its only foreseeable risk is misuse of the retained numbers outside their recorded scope, which Section 7 exists to constrain. The withdrawal of an earlier interpretation is reported in full rather than silently corrected.

B MACHINE-CHECK TRANSCRIPT

Every scanner shipped with this note, with the exit code it returns on the frozen text at compile time (Section 3.2). Paths are relative to the release root, and each scanner’s path heads its own full-width row so that no path is broken across lines.

C SCANNER VALIDITY: INDEPENDENT RECOVERY AND COVERAGE SELF-CRITIQUE

Two 0-GPU validity studies (Section 8) bound how the six-family scanner may be read.

Table 5: Machine-check transcript: for each scanner, its target, its exit code on the frozen text, the counts it emits, and its red-team verdict. A fixture that must *fail* is what separates a check with discriminative power from one that always passes; n/a marks a structural or same-source check for which a fixture is not the relevant evidence.

Target	Exit	Match counts (frozen text)	Red-team verdict
bundle/scanners/audit_p0_1_critical_path_scope.py main.tex; G1-G4 scope	0	G1=4, G2=3, G3=0, G4=0	redteam/FIXTURE1 fails (rc=1): G4 rejects concurrency-throughput gloss
bundle/scanners/audit_grad_mode_guard.py vendored e2e bench sources	0	baseline guarded, worker unguarded; fix locus present	n/a (structural)
bundle/scanners/verify_p0_2_adapter_provenance.py bundle; FACT-A/B	0	FACT-A 24/24 manifest OK; FACT-B 0 weight files	n/a
tools/register_ledger_check.py registry LEDGER/REGISTER	0	12 rows exact-set; statuses canonical	9 known-bad mutations all <i>fail</i>
tools/gen_register_tables.py main.tex Table 2 (same source)	0	12 rows = registry, no drift	n/a (drift vs checker = build fail)
bundle/method/independent_claim_audit.py main.tex, six families	0	68 hits (28 WITHDRAWN / 8 NOT_MEASURED / 5 OVERCLAIM_RISK / 5 HEDGE / 19 PROVENANCE / 3 MATCHED)	independent manuscript (S1) 91 hits, distinct failure-mode structure

Independent-annotator recovery. A reviewer blind to our ledger re-extracted the load-bearing claims from the frozen PDF and recovered all 14 C/D/S register IDs with the same canonical semantics the paper assigns them (51 claims: 21 MEASURED, 10 NOT_MEASURED, 7 withdrawal, 5 scope, 5 method-instrument, 3 cross-case; bundle/validity/annotator_recovery.json). Its skeptical omissions were all already-honest disclaimers in the paper (single-point D1 replication; the author-constructed ledger limit; inspection-grade 16k attribution); none was a registered claim absent from the paper.

Coverage self-critique. A blind mapping of this note’s own prose honesty onto the six families left 9 of 27 structures outside any family — the structures that most distinguish the note: an *attribution-grade* distinction (measured vs. *by inspection* vs. *analytical-only* vs. derived), declared-universe completeness, refusal-to-aggregate integrity, effective-vs-nominal sample size, and false-precision (bundle/validity/coverage_self_critique.json). The six families are a *necessary* hygiene filter: a scanner built purely from them would pass this note’s strongest honesty structures unseen.

D Q^* RECONCILIATION AND THE DERIVED SEQUENCES

The two tables the C13 and C7 estimators of Section 5 point at. Both are read as rows rather than as prose because every entry is one of a set of comparable quantities.

E INTERVAL ESTIMATES FOR C9 AND C13

The two intervals this note cites, each drawn from its own resampling population, as referenced by the C9 estimator of Section 5.

Table 6: C13 Q^* reconciliation at $L=128k$. The inputs are the printed per-query components of `bundle/pl_8_serving/pl_8_serving_aggregate.json` (cell gpu); the one-time numerator $W_{\text{CoMem}} - I_{j0}$ is shared by every row, so the rows differ only in their denominator. Each citable row reproduces from the printed inputs. The superseded face is a replicate-level paired-bootstrap reanalysis (`bundle/qstar_dispersion_reanalysis.json`): its denominators are bootstrap means over replicates and are *not* recomputable from the printed per-query components, which is why the aggregate grid supersedes it for point values, and its own datacard flags it as not a reproduction of the frozen $G=1$ figure.

Case	$W_{\text{CoMem}} - I_{j0}$ (s)	mean paired margin (s/q)	Q^*	denominator recomputable
<i>aggregate grid</i> — the citable face				
$G=1$	6.653−0.022	0.9379−0.6807 = 0.2572	25.8	yes
$G=512$	6.653−0.022	20.9617−20.9213 = 0.0404	164.2	yes
<i>superseded face</i> — replicate-level paired-bootstrap reanalysis				
$G=1$	6.653−0.022	0.2430 (reanalysis)	27.3	no
$G=512$	6.653−0.022	0.0536 (reanalysis)	123.6	no

Table 7: Derived sequences behind C7 and S , moved out of prose so they are read and compared as comparable rows. The S row is an algorithmic ESTIMATE on representative chunks (64/256/2048 at 32k/128k/1M), not a frozen-corpus measurement.

ID	Quantity	Value	Estimand	Scope
C7	prefill-time growth in L	1.46	serial fit, 5 points	not a memory slope
C7	peak-memory growth in L	0.48	serial fit, 5 points	not a time slope
C7	peak-memory rate in L	0.438 GB per 1k	serial, near-affine	—
S	per-query chunk scoring	0.0038/0.0142/ 0.1255 s/q	algorithmic estimate	32k/128k/1M

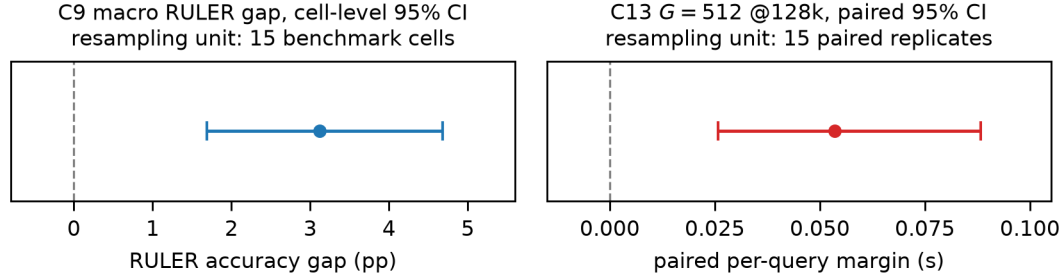


Figure 3: The two interval estimates this note cites, each with its own resampling population. **Left:** the C9 macro RULER gap, cell-level 95% percentile bootstrap [1.69, 4.68] pp (15 cells, seed 123, $n_{\text{boot}}=20000$). **Right:** the *dispersion* of the C13 $G=512$ @128k per-query margin, paired bootstrap [0.026, 0.088] s (15 units). The right panel is a dispersion measure; the citable point margin is the recomputable aggregate figure 0.0404 s/q (Table 3 C13). Neither interval includes zero; the axes differ because the quantities differ.

F FULL NOTATION AND LEDGER-ID GLOSSARY

The row definitions that were removed from the main-text notation table (Table 1) to meet the page budget are collected here for completeness; they are the same definitions the archive used.

Table 8: Full notation and internal identifiers. Identifier families (C, D, S, R) index the archived claim ledger. Rows are grouped as terms, then identifier families.

Term	Definition
CoMem	The archived cached-memory serving system: writes one long document’s layer- j hidden states once, then replays layers $j \dots 36$ per query instead of re-prefilling.
QCMem	The vendored selector module inside the CoMem code base; source of the <code>create_causal_mask</code> argument mismatch of Section 6.
$j0$ ($j=0$)	The <i>full-depth</i> arm: nothing cached; selected raw token-id chunks replay through all 36 layers, so its per-query read is the most expensive.
$j12$	The flagship <i>cached</i> arm ($j=12$ hidden states plus a LoRA adapter (Hu et al., 2022)); replays 24 layers.
B-only	The full-context (dense) baseline measured <i>without</i> the LoRA adapter (arm B); the matched adapter arm (A) was unrecoverable, so every staircase cell is B-only.
S	Per-query CPU cost of the chunk- <i>selection</i> step (iterated BM25 scoring); an algorithmic estimate, never measured on the frozen corpus.
G	Concurrency (in-flight requests), or the generated sequence length in the archived decode model, as labelled per row.
Q^*	Break-even query count at which the cached arm’s amortized cost falls below the full-depth arm’s; defined by Eq. 1.
C4, C7, C9, C13	Stable claim IDs of the archived ledger (serving cost; capacity staircase; RULER quality; crossover), so each frozen value maps to its record.
D1–D6	The six decisive-evidence debts recorded at closure and their scope: D1 concurrent full-context serving @ $L=32k$ (measured fix-and-rerun), D1-16k the same @ $L=16k$ (unmeasured re-run), D2 adapter-source recovery, D3 KV-selector ($j0$) finite fetch, D4 adapter-identity table completeness, D5 adapter-identity per-row binding caveat, D6 LongEval per-item re-run. Single-arm isolation is class S1 (below), not D5.
S1, S3, S6	Three soundness-deficit classes from the review taxonomy: S1 the titled mechanism was never isolated by a single arm; S3 $n=1$ treated as an effect; S6 no control-family member was run.
R142	The internal round at which the flagship $j=12$ adapter arm was withdrawn; recorded in later run configurations as <code>adapter=NONE_WITHDRAWN_R142</code> .
“0 data”	A GPU burn-ledger column counting <i>usable measurement records</i> from one job; “0 data across five fires” means five GPU jobs were charged and none produced one.